Davide Macario

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Skills

Technical	Python, Golang, Terraform, AWS, Linux, Bash, Kubernetes, Docker, CI-CD tooling (Git, GitHub Workflows, GitLab CI), Zero-trust, Ansible, Monitoring (Prometheus, Loki, Grafana, OpenSearch).
Languages	Italian (native), English (C1 – IELTS 8.0), French (basic), Dutch (basic).
Interests	ML/AI applications, Open-Source Software, Linux, MLOps, Self-Hosting/Homelabbing, Distributed Computing, Software Architectures, Edge Computing, IoT.

Experience

Vitestro B.V., *DevOps Engineer* | Utrecht, NL — Mar 2026 – Current

- Lead the effort to achieve integration of Aletta™ with customer hospitals' Laboratory Information Systems (LIS); deployed an integration engine hosted on **AWS (ECS Fargate)**, including centralized monitoring and alerting platform.
- Implemented full solution for **remote data upload** (metrics, logs, venipuncture session data) from the device to the cloud, consisting **Fluent Bit** and a custom **Python** application running on the device, pushing data to **AWS S3** and **OpenSearch**.
- Lead the development of the **internal Cloud Platform** to enable other R&D teams to easily deploy applications to AWS; solution covers full lifecycle of application on the cloud, from CI (**GitLab CI**), to CD (**Terraform** and AWS ECR), to monitoring and alerting (**Amazon-managed Prometheus**, Alertmanager, Grafana).
- Supported other R&D teams with **internal tooling**, including data migrations, package registries, container registries, container image development.

Vitestro B.V., *Entry-Level DevOps Engineer* | Utrecht, NL — Sep 2024 – Feb 2026

- Member of *Platform & Connectivity* team working on integrating the world's first **Autonomous Robotic Phlebotomy Device** – **Aletta™** with the cloud platform hosted on **AWS**.
- Designed and implemented full **GitLab CI** ecosystem on **AWS**, including autoscaled runners, Docker registry mirror, repository caches, embedded Linux builds (Yocto, Bitbake); achieved costs reduction of 20% compared to preexisting setup, and enabled scaling to support the growing number of active developers in the organization.

University of Illinois at Chicago, *Graduate Research Assistant* | Illinois, USA — Jan 2024 – Jun 2024

- Research work focusing on distributed computing applied to **Machine Learning** and **Generative AI** models.
- Designed and implemented a framework for deploying state-of-the-art LLMs on networks of constrained edge devices.
- Won **Best Paper Award** at IEEE LANMAN 2025 for the work on *"A Model-Distributed Inference Approach for Large Language Models at the Edge"*.

Education

University of Illinois at Chicago, *MS in Electrical and Computer Engineering* | IL, USA — Jul 2024

Double-degree program in collaboration between Politecnico di Torino and University of Illinois at Chicago.

Politecnico di Torino, *MS in ICT for Smart Societies (Computer Engineering)* | Torino, Italy — Jul 2024

M.Sc. Thesis: "A Model-Distributed Inference Approach for Large Language Models at the Edge".

Politecnico di Torino, *BS in Electronics and Communications Engineering* | Torino, Italy — Sep 2022

Associations: IEEE Eta-Kappa-Nu Mu-Nu chapter member (honor society – Politecnico di Torino).

Projects

DMHosted – Homelab and Self-Hosting — Nov 2025 – Current

- Bare-metal **Kubernetes** (K3s) cluster, running both internal (VPN) and public-facing services.
- **GitOps** setup via ArgoCD, centralized **monitoring and observability** (Prometheus), **Ansible** for provisioning and server upgrades, **CloudNativePG** for databases, **Tailscale** as VPN, and **Cloudflare Tunnels** for public services.
- Includes a handful of Linux servers connected over VPN (Tailscale), running services via Docker to complement main K8s cluster (incl. storage, backups, monitoring, GPU compute).
- `https://github.com/davmacario/dmhosted-infra`

A Model-Distributed Inference for Large Language Models at the Edge | M.Sc. Thesis — Jan 2024 – Jun 2024

- Implementation of **pipeline parallelism** for **Large Language Models**, allowing to run LLM inference on a network of edge devices (Nvidia Jetson TX2).
- Employing **distributed computing** techniques to achieve increased generation rates by minimizing idle nodes when serving multiple user queries at the same time.
- Ability to run the **GPT-2**, **Llama 3**, and **Mistral** model families achieving usable throughput ($>$ 15 tokens per second).
- `https://github.com/davmacario/MDI-LLM`

FREISA | OpenCV AI Competition 2023 (*Grand prize winner*) Jul 2023 – Dec 2023

- *"Four-legged Robot Ensuring Intelligent Sprinkler Automation"*.
- Computer vision-enabled quadruped robot able to detect plants, approach them and decide whether to sprinkle water on them based on the condition of the leaves.
- Focused on developing the computer vision module using an OAK-D Lite camera, **YOLOv8**, **OpenCV**, and **DepthAI**, providing fast APIs to the motion controller (ROS 2), and running fully on-board.
- `https://github.com/B-AROL-O/FREISA`

Certifications

Software Architecture for Developers – by Simon Brown (Trifork) May 2026
HashiCorp: Terraform Associate (003) Aug 2025

Achievements

Best Paper Award – IEEE LANMAN 2025, Lille, France Jul 2025
Alta Scuola Politecnica (ASP) – honor programme for top 60 students during M.Sc. Sep 2022 – Jul 2024
TOP-UIC Scholarship – awarded on academic performance for the double-degree programme May 2023
Percorso Giovani Talenti – honor programme for top students during B.Sc. Sep 2019 – Jul 2022